

Ahmed Mohamed

AI Engineer
Technical Lead

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- 📍 Melbourne, Australia

Education

- PhD, Bioinformatics & Chemical Genomics** 2012–2016
Kyoto University, Japan
- MSc, Advanced Pharmacology-science** 2010–2012
Osaka University, Japan
- BPharm, Pharmaceutical Sciences** 2002–2007
Suez Canal University, Egypt

Skills

- LLM engineering**
 - Extraction pipelines, structured output
 - Cell-level evals, DSPy prompt optimisation
 - RAG, multi-provider tooling
- Data science**
 - Statistics & ML at population scale
 - Cancer genomics, Bioconductor
 - lipidr, NetPathMiner, vissE (author)
- Software & infra**
 - Python, Go, TypeScript/React
 - AWS, Kubernetes, Docker, PostgreSQL
 - Nextflow, SLURM HPC
- Web3**
 - Ethereum staking services
 - Financial reporting & invoicing

Awards

- ICR Award for Graduate Students, 2015
- Winner, Novartis BioCamp, 2013
- JSPS International Training Program Fellowship, 2013
- Rotary Yoneyama Memorial Scholarship

Publications

30+ papers. NetPathMiner (Bioinformatics), NMRPro (Bioinformatics), lipidr (160+ citations, in Bioconductor since 2019), vissE.cloud (NAR, first author).

Languages

English, Arabic, Japanese

I build LLM systems that go into production and stay there: extraction pipelines, multi-provider structured output, cell-level accuracy evaluation, and the AWS and data-governance setup to run them on sensitive data. Data scientist by training (PhD, 30+ papers, statistics and ML on a 30,000-sample study) and a backend engineer by practice (Go services and financial reporting for an institutional Ethereum operator).

Professional Experience

- Co-founder & Technical Lead** AIWhisperers, Australia 2023 – now
 - Built production LLM systems for SMEs, including a human-reviewed platform converting supplier contracts into structured commercial data for a B2B travel company.
 - Wrote the company's LLM tooling: multi-provider structured-output library, composable extraction chains, LLM-run caching, cell-level evaluation comparing prompt strategies and models; DSPy prompt optimisation.
 - Established the PHI data-governance framework (Privacy Act 1988, ISO 27001-aligned) and secure AWS architecture: private-subnet SageMaker inference, KMS-encrypted RDS/S3, least-privilege IAM.
 - Delivered LLM-for-business workshops for companies in retail, health, tourism and mining.

LLM systems Evals Data governance AWS
- Senior Software Engineer** Bitwise Onchain Solutions (ex-Attestant) 2022 – now
 - Backend engineer for institutional Ethereum and multi-chain staking services.
 - Own the client reporting and invoicing system with multi-currency support; maintain client-facing APIs and the multi-chain staking dApp.

Go Ethereum Financial systems
- Bioinformatician, Senior Research Officer** WEHI, Melbourne 2020 – 2025
 - Led experimental design, QC, statistics and ML for the ASPREE proteomics programme: 12,138 participants, ~30,000 LC-MS runs, >10 TB — at the time 10× the largest published MS population proteomics study.
 - Built the Nextflow processing pipeline on SLURM HPC (DIA-NN, out-of-memory HDF5 matrices) and a BigQuery-backed QC dashboard watching hundreds of runs a day.
 - Designed the analysis-first experimental layout that made batch correction and longitudinal modelling work at scale; ran LASSO biomarker models across 13 conditions.
 - Built vissE.cloud end-to-end (React/TypeScript, Flask, Redis, Docker): a public gene-set analysis web server across bulk, single-cell and spatial data; first author of the NAR paper.
 - Senior author on an ML urine-spectroscopy test for bladder-cancer recurrence (ROC > 0.8).

Proteomics Statistics & ML Nextflow BigQuery
- Research Officer** QIMR Berghofer & UQ Diamantina, Brisbane 2017 – 2020
 - Created lipidr, an R/Bioconductor package for lipidomics analysis (160+ citations), and Lipid-Suite, its web server for non-programmers; sole developer of RaftProt V2 — tools accessed tens of thousands of times.
 - Statistical lead for collaborator studies; taught R package development and Software Carpentry Python; associate supervisor of a PhD student to completion.

Lipidomics Bioconductor Biostatistics
- PhD · JSPS Visiting Fellow** Kyoto University · Boston University 2012 – 2016
 - NetPathMiner, an R/Bioconductor package for network path mining (Bioinformatics); NMRPro, interactive web processing of NMR spectra (Bioinformatics).
 - Pan-cancer analysis of somatic mutations in protein complexes.

Network biology Cancer genomics
- Tailor** This CV — career knowledge base + agentic tailoring, open source 2026
 - Harvest your career once; tailor CVs to any job in minutes. This document is its output.

Claude Code typst
- Notebrainer** Provenance-first agent memory · Claude Code plugin 2025
 - Memory layer where every stored fact traces to its source text.

Postgres pgvector
- MapMyMath** Procedural maths game · web, iOS, Android 2024
 - 1,320 catalogued exam questions; built solo in five weeks.

Python React/WebGL